Does Adaptive Resampler Selection Help Modern Tree Ensembles? ARS-Stack and a Controlled Study on 28 Imbalanced Datasets

[Author Name][1] and [Co-author Name][1]

[1][Department, Institution, City, Country], [email@institution.edu]

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Abstract

Class imbalance is usually handled by resampling the training data and then fitting an ensemble. Most hybrid pipelines apply one resampler to the whole minority class and combine their learners with fixed weights. We propose ARS-Stack (Adaptive Resampler-Selection Stacking), which clusters the minority class, scores each cluster for class overlap and local density, selects a resampler *per cluster*, and combines Random Forest, XGBoost and LightGBM with confidence weights. We evaluate it under a leakage-free, budget-matched protocol on 28 datasets (credit-card fraud, mammography, 10-class yeast and 25 benchmarks) against eight baselines and ablations: 3,780 cross-validation folds, analysed with frequentist and Bayesian tests. ARS-Stack is significantly and practically better than global SMOTE-ENN (Bayesian probability ≥ 0.96 on F1, AUC-PR and MCC). It is, however, *practically equivalent* to plain SMOTE, SMOTE-Tomek and its own ablations, and it cannot be separated from training without resampling. Diagnostics explain why: the confidence weights are almost uniform, Tomek cleaning removes nothing after 1:1 SMOTE, and the selector rarely varies its choice within a dataset. A pilot of validated selection closes the gap to the baselines but does not exceed them. Resampling helps mainly when the minority class is very small (mean AUC-PR gain $+0.012$ below 75 minority samples, -0.009 above). We release the reproducible pipeline and discuss five evaluation pitfalls that make new resamplers look better than they are.

Keywords: class imbalance, resampling, SMOTE, ensemble learning, stacking, statistical comparison of classifiers, Bayesian analysis, MLOps.

1 Introduction

In fraud detection, medical screening, intrusion detection and churn prediction, the class of interest is usually the rarest. A classifier that minimises overall error can then reach high accuracy while missing most minority cases, which keeps class imbalance a central problem in applied machine learning [1–4]. The standard remedies are *data-level* resampling, which generates synthetic minority samples or removes majority ones [5–7], and ensembles that reduce the variance and bias of minority predictions [8–10]. Hybrid pipelines that resample the data and then fit an ensemble dominate recent applied work, particularly on credit-card fraud [11–13].

Our literature review (Section 2) found three recurring gaps. First, one resampler is chosen by hand and applied *uniformly* to the whole minority class, although the best resampler depends on the dataset [14, 15] and plausibly on the region: small disjuncts [16], borderline points [17] and overlap zones [18] each call for different treatment. Second, ensemble combiners are static and trust every learner equally [19]. Third, multiclass imbalance is under-studied [20, 21].

We designed ARS-Stack to address all three. It clusters each minority class, scores the clusters for overlap and density, picks a resampler per cluster, and combines Random Forest [22], XGBoost [23] and LightGBM [24] with confidence weights. It supports binary and multiclass problems and ships as a reproducible MLOps system.

Evaluating it fairly proved harder than building it. We found, and removed from our own pipeline, five pitfalls that each manufacture a spurious improvement: unequal oversampling budgets, confidence

weights cancelled by a meta-learner, rank-forced selection rules, fixed decision thresholds, and duplicate records that straddle train and test splits. With them removed, we asked whether adaptive resampler selection helps a modern tree ensemble at all. The answer is largely negative, and we argue that this answer is the more useful result.

Contributions.

1. **Method:** ARS-Stack, with scale-free cluster scores, a global synthetic-sample budget, confidence-weighted averaging with a tuned threshold, and a multiclass extension (Section 3).
2. **Protocol:** a leakage-free, budget-matched evaluation in which scaling, resampling, weighting and thresholds are fitted inside each fold (Section 4).
3. **Study:** nine variants on 28 datasets with repeated cross-validation, analysed with Friedman, Nemenyi, Holm-corrected Wilcoxon and corrected resampled t-tests, and with Bayesian tests using a region of practical equivalence [25] (Section 5).
4. **Diagnosis:** why each component does or does not help, a validated-selection pilot, and a meta-analysis of when resampling helps (Section 6).
5. **Artifact:** an open pipeline (DVC, MLflow [26], Prefect, Docker, CI, FastAPI) that records code version, packages and hardware with every result.

In short, ARS-Stack beats aggressive cleaning (SMOTE-ENN) but is practically equivalent to plain SMOTE and to its ablations, and it cannot be separated from no resampling. This agrees with growing evidence that strong learners gain little from oversampling [27–29].

2 Related Work and Research Gaps

Resampling. SMOTE [5] interpolates between a minority sample and one of its k nearest minority neighbours and remains the default oversampler [30]. Its descendants mainly change *where* samples are placed. Borderline-SMOTE [17] and ADASYN [6] focus on hard points; Safe-Level SMOTE [31] and MWMOTE [32] avoid noisy regions; k-means SMOTE [33] generates only inside safe clusters. Cleaning methods remove ambiguous points: Tomek links [34] delete cross-class nearest-neighbour pairs, and ENN [35] removes points misclassified by their neighbours. Both were combined with SMOTE by Batista *et al.* [7]. In Kovács's comparison of 85 oversamplers on 104 datasets [15], no method dominates.

Cluster- and region-aware resampling. The minority class is often a union of sub-concepts, and errors concentrate in small disjuncts [16]. Cluster-based methods [16, 33] and DCSHS [18], which uses dual clustering and stage-wise hybrid sampling in overlap zones, adapt *where* samples are generated. They still apply one fixed *technique*, however. We found no method that chooses among several resamplers per region.

Ensembles. Boosting (SMOTEBoost [9], RUSBoost [36]), bagging (EasyEnsemble [10]) and stacking [37] have all been adapted to imbalance [8]. Random Forests and gradient-boosted trees [22–24] are now the usual base learners in hybrid fraud pipelines [12, 13]. Instance-adaptive combination is studied as dynamic classifier selection [38]. Resampling pipelines, however, typically combine their learners with a fixed vote or an unweighted meta-learner [19].

Evaluation. Under imbalance, precision–recall metrics [39], the G-mean [40] and the MCC [41] replace accuracy. Across datasets, Demšar [42] recommends the Friedman test [43, 44] with post-hoc tests [45, 46]. Within a dataset, correlated folds call for the corrected resampled t-test [47, 48]. Bayesian tests with a region of practical equivalence (ROPE) [25, 49, 50] can also support *equivalence*. Several studies doubt that resampling helps strong learners: gains vanish with boosted trees and tuned thresholds [27], imbalance corrections harm calibration [28], and benefits depend on the learner [29].

Research gaps. Table 1 condenses the ten studies of our review into four gaps: **G1**, no resampler dominates, yet pipelines fix one globally; **G2**, ensemble weights are static; **G3**, multiclass imbalance is under-explored; **G4**, evidence comes from few datasets and frequentist tests, and rarely controls the amount of resampling or leakage from resampling before splitting. ARS-Stack targets G1–G3; our protocol targets G4.

Table 1: Studies from our literature review and the gaps they leave open (mapped to G1–G4).

Ref.	Approach	Data	Limitation / gap
SHREA [51]	Oversample-impute, undersample, one-class classifiers, bagging	Generic benchmarks	No large fraud/medical test; no SMOTE-ENN comparison (G4)
DCSHS [18]	Dual clustering, stage-wise hybrid sampling in overlap zones	Public imbalanced sets	Region-aware but one sampling scheme (G1)
[20]	Survey: resampling + ensembles, binary vs. multiclass	Health, finance, security	18/24 prior works binary only (G3)
[52]	Gradient-based instance-level resampling	Cost-sensitive benchmarks	Tied to boosting internals, hard to reproduce (G4)
[19]	Spatial-distribution undersampling + voting	Various	Static voting weights (G2)
[21]	SMOTE + Isolation-Forest filtering	Medical multiclass	Medical only; no stacking (G1, G3)
[11]	EasyEnsemble, XGBoost, RF, AdaBoost	Fraud, phishing	Benchmarks existing methods only (G1)
[14]	ROS/RUS/hybrid × bagging/boosting	Water quality	Best resampler dataset-dependent, no selector (G1)
[12]	SMOTE + RF/XGBoost/LightGBM/stacking	IEEE-CIS, fraud	Resampler fixed to SMOTE (G1)
[13]	SMOTE/SMOTE-Tomek + stacking	Credit-card fraud	One domain, manual resampler (G1, G4)

3 The ARS-Stack Method

Figure 1 shows the system. The data layer versions and cleans the data. The adaptive-resampling layer, the methodological core, prepares the training set for the ensemble layer. An MLOps layer makes every run reproducible. We describe the binary case (minority class c) first and then the multiclass extension.

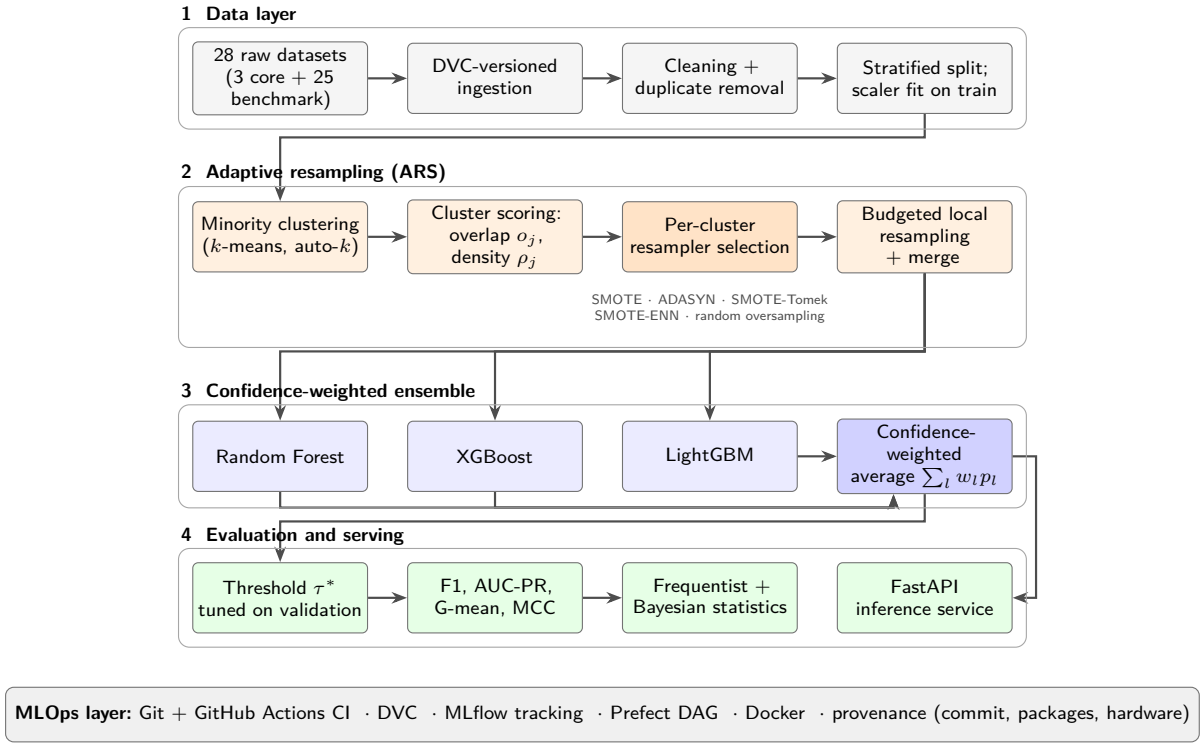


Figure 1: ARS-Stack architecture. The minority class is clustered, each cluster is scored and resampled with its own resampler under a global synthetic-sample budget, and three tree ensembles are combined with confidence weights and a validation-tuned threshold. An MLOps layer handles versioning, tracking, orchestration, containers and serving.

Minority clustering. The minority samples M of the training split are partitioned with k-means [53], choosing $k \in \{2, \ldots, 6\}$ by silhouette score [54]. Clusters with fewer than 10 samples are merged into the largest, and a class with fewer than 20 samples is kept as one cluster. DBSCAN [55] is evaluated as an ablation.

Cluster scoring. The *overlap score* of cluster C_j is the mean share of majority samples among each point's $k_o = 10$ nearest neighbours in the full training set:

$$o_j = \frac{1}{|C_j|} \sum_{x \in C_j} \frac{1}{k_o} \sum_{x' \in \mathcal{N}_{k_o}(x)} \mathbb{1}\left[y(x') \neq c\right]. \tag{1}$$

With $\delta(C)$ the inverse mean distance of the points of C to their $k_d = 5$ nearest neighbours within C, the *density ratio* $\rho_j = \delta(C_j)/\delta(M)$ compares the cluster with its whole class. Unlike raw distances, it does not depend on the feature scale.

Rule-based selection. Each cluster receives a resampler through rules fixed before any experiment:

$$r_j = \begin{cases} \text{random oversampling} & |C_j| < 6, \\ \text{SMOTE-ENN} & o_j \geq 0.7, \\ \text{SMOTE-Tomek} & 0.4 \leq o_j < 0.7, \\ \text{ADASYN} & \rho_j \leq 0.75, \\ \text{SMOTE} & \text{otherwise.} \end{cases} \tag{2}$$

The rules are checked top to bottom. Strong overlap ($o_j \geq 0.7$: most neighbours belong to the majority class) calls for aggressive ENN cleaning, boundary overlap for lighter Tomek-link cleaning [7], and a cluster markedly sparser than its class ($\rho_j \leq 0.75$) for ADASYN's density-adaptive generation [6]. Well-separated, typically dense clusters get plain SMOTE. A percentile-based variant of the rules is used only with at least three clusters: with one or two clusters, percentiles decide by rank alone.

Budgeted local resampling. To compare resamplers rather than resampling amounts, every variant adds the same number of synthetic samples, $B = \max(0, \text{round}(\beta n_{\bar{c}}) - n_c)$ with $\beta = 1$. ARS-Stack splits B across clusters in proportion to their size, $b_j \propto |C_j|$. Each cluster is resampled inside its neighbourhood, defined as the cluster plus its $3|C_j|$ nearest majority points, with r_j producing $|C_j| + b_j$ minority samples. All synthetic samples are then merged. A majority point counts as removed only if every cleaning step that saw it removed it. If a resampler fails, SMOTE fills its allocation.

Confidence-weighted ensemble. Random Forest, XGBoost and LightGBM (300 trees each) are trained on the resampled data and weighted by their validation AUC-PR:

$$w_l = \frac{\max(\text{AP}_l^{\text{val}}, 0.05)}{\sum_{l'} \max(\text{AP}_{l'}^{\text{val}}, 0.05)}, \qquad \hat{p}(x) = \sum_l w_l \, p_l(x). \tag{3}$$

The decision threshold τ^* maximises F1 on the validation split. The weights are applied in an *average* on purpose: fed as inputs to a logistic-regression or tree meta-learner, they would be cancelled, because the meta-learner rescales its inputs. Classic stacking [37] is evaluated as an ablation.

Multiclass extension. Every class except the largest is clustered, scored and resampled in turn, up to β times the largest class. The weights use macro-averaged AUC-PR, and the prediction is the arg-max. Algorithm 1 summarises training.

Implementation. The pipeline uses scikit-learn [56], imbalanced-learn [57], XGBoost and LightGBM. Data are versioned with DVC, runs are tracked in MLflow [26] and orchestrated as Prefect tasks, the environment is fixed in Docker, and CI runs over 120 unit tests. A FastAPI service serves the models. Every result records the git commit, package versions and hardware, and saved folds are keyed by a fingerprint of the configuration and code, so results from different code versions cannot mix.

Algorithm 1 ARS-Stack training on one training split

Require: training data (X, y), validation data (X_v, y_v), ratio β

for each non-majority class c **do**
  $\{C_j\} \leftarrow \textsc{Cluster}(\{x_i : y_i = c\})$; compute o_j, ρ_j
  $B \leftarrow \max(0, \text{round}(\beta\, n_{\max}) - n_c)$; $b_j \propto |C_j|$
  for each cluster C_j **do**
    $r_j \leftarrow$ Eq. 2; $N_j \leftarrow C_j \cup$ nearest $3|C_j|$ majority points
    $S_j \leftarrow r_j(N_j)$ with target $|C_j| + b_j$ ▷ fallback: SMOTE
  end for
  merge synthetic samples; drop majority points removed in all contexts
end for
fit RF, XGBoost, LightGBM; $w_l \leftarrow$ Eq. 3; $\tau^* \leftarrow \arg\max_\tau \text{F1}^{\text{val}}$
return $\hat{p} = \sum_l w_l p_l$ and τ^*

4 Experimental Setup

Datasets. We use 28 datasets, listed with their characteristics in Table 6. The three *core* datasets cover the imbalance settings of our project plan:

- credit-card fraud (`creditcard` [58]): binary, imbalance ratio (IR) 599;
- mammography [59]: binary, IR 30;
- yeast [60]: kept as a 10-class problem, smallest class 5 samples.

The 25 *benchmarks* are the whole imbalanced-learn collection [57, 61] except the two sets that overlap the core datasets; none was excluded for size. Together they span 330–283,726 samples, 6–617 features and IR 6–599. Exact duplicates are removed before splitting (1,081 rows from creditcard, 30% of mammography).

Variants. Table 2 lists the nine variants. All share the base learners, hyperparameters, weights and threshold tuning. All except *No resampling* target the same class balance ($\beta = 1$), so each comparison with ARS-Stack isolates one design decision.

Table 2: Compared variants, their code names (used in Figures 3 and 4), and what each comparison with ARS-Stack isolates.

Variant (code name in figures)	Description (isolated decision)
ARS-Stack (`ars_stack`)	Per-cluster rule-based selection + confidence-weighted average
Equal-weight avg. (`equal_average`)	Equal weights (value of confidence weighting)
Meta-learner stacking (`meta_stacking`)	Logistic-regression stacking on out-of-fold predictions
ARS (DBSCAN) (`ars_dbscan`)	DBSCAN clusters (clustering robustness)
SMOTE, ADASYN (`plain_smote`, `plain_adasyn`)	One global resampler (value of adaptive selection)
SMOTE-Tomek, SMOTE-ENN (`plain_smote_tomek`, `plain_smote_enn`)	Global SMOTE with Tomek-link / ENN cleaning
No resampling (`no_resampling`)	Same ensemble without resampling

Protocol. Each dataset is evaluated with stratified 5-fold cross-validation repeated three times (seed 42), giving 15 paired folds per dataset and variant (Figure 2). Within each fold, the pipeline fits every component on the outer training part only:

- a stratified 15% validation split is held out;
- the scaler is fitted and resampling applied on the remaining training data;
- the confidence weights and threshold come from the validation split;
- the outer test fold is used once, for scoring.

Resampling therefore never touches evaluation data.

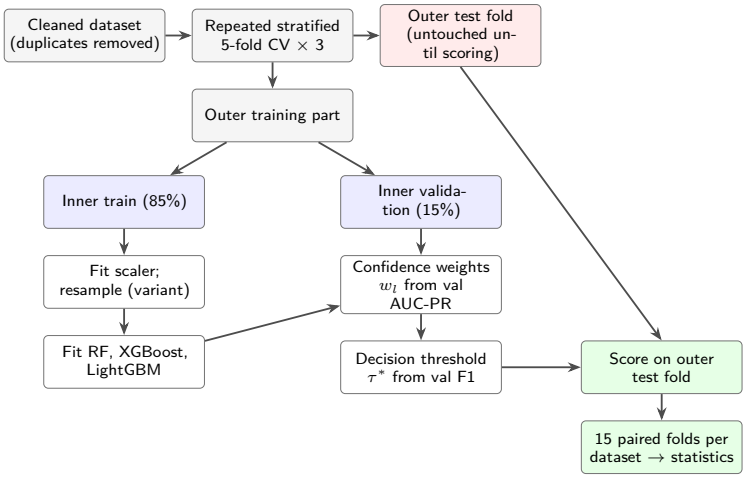


Figure 2: Leakage-free protocol: scaling, resampling, weights and threshold are fitted inside the outer training part of each fold.

Learners and metrics. Hyperparameters were fixed in advance and are shared by all variants:

- Random Forest: 300 unpruned trees;
- XGBoost: 300 trees, depth 6, learning rate 0.1, row and column subsampling 0.8;
- LightGBM: 300 trees, learning rate 0.1;
- all resamplers: $k = 5$ neighbours.

We report minority-class F1, AUC-PR (average precision, our primary threshold-free metric), G-mean and MCC [39–41]. For yeast, F1, AUC-PR and G-mean are macro-averaged.

Statistics.

- **Across datasets** [42]: average ranks, the Friedman and Iman–Davenport tests [43, 44], the Nemenyi critical difference, and Wilcoxon tests of ARS-Stack against each variant [62], Holm-corrected [46].
- **Within datasets**: the corrected resampled t-test [47, 48], because the 15 folds are correlated.
- **Bayesian**: correlated t-tests per dataset [49] and signed-rank tests across datasets [50], with a ROPE of ± 0.01 and a decision when one region reaches posterior probability 0.95 [25].

Computation. The $28 \times 9 \times 15 = 3{,}780$ folds took 16.2 compute-hours on a 16-thread CPU with an RTX 3050 GPU (used by XGBoost), from a single committed code version.

5 Results

Overall ranking. Table 3 and Figure 3 give average ranks over the 28 datasets. The Friedman test rejects equal performance for F1 ($p = 0.0018$), AUC-PR ($p = 1.4 \times 10^{-4}$) and MCC ($p = 5.8 \times 10^{-4}$), but not for G-mean ($p = 0.11$). The differences come almost entirely from the two most aggressive global resamplers, SMOTE-ENN (AUC-PR rank 7.11) and ADASYN (6.16). The other seven variants lie within about one rank of each other. ARS-Stack ranks 5th on AUC-PR (4.70), 4th on G-mean and 7th on F1 and MCC, while training *without* resampling ranks first on AUC-PR (4.21) and MCC (4.29).

Critical differences. With 9 methods and 28 datasets, the Nemenyi critical difference is 2.27 ranks (Figure 4). For AUC-PR, SMOTE-ENN is significantly worse than every method ranked 4.84 or better, including ARS-Stack. For F1, it is significantly worse only than ARS with DBSCAN (4.20) and SMOTE-Tomek (4.25), and ARS-Stack (4.79) is not separated from it. No other pair of methods differs significantly.

Table 3: Average rank over 28 datasets (1 = best; best in bold) and Friedman p-values.

Variant	F1	AUC-PR	G-mean	MCC
No resampling	4.68	**4.21**	4.96	**4.29**
SMOTE	4.68	4.23	4.93	4.64
ARS (DBSCAN)	**4.20**	4.29	4.52	4.55
SMOTE-Tomek	4.25	4.30	**4.39**	4.36
ARS-Stack	4.79	4.70	4.61	4.82
Equal-weight avg.	4.79	4.71	4.46	4.64
Meta-learner stacking	4.55	5.29	5.02	4.45
ADASYN	6.32	6.16	6.36	6.25
SMOTE-ENN	6.75	7.11	5.75	7.00
Friedman p	0.0018	0.00014	0.11	0.00058

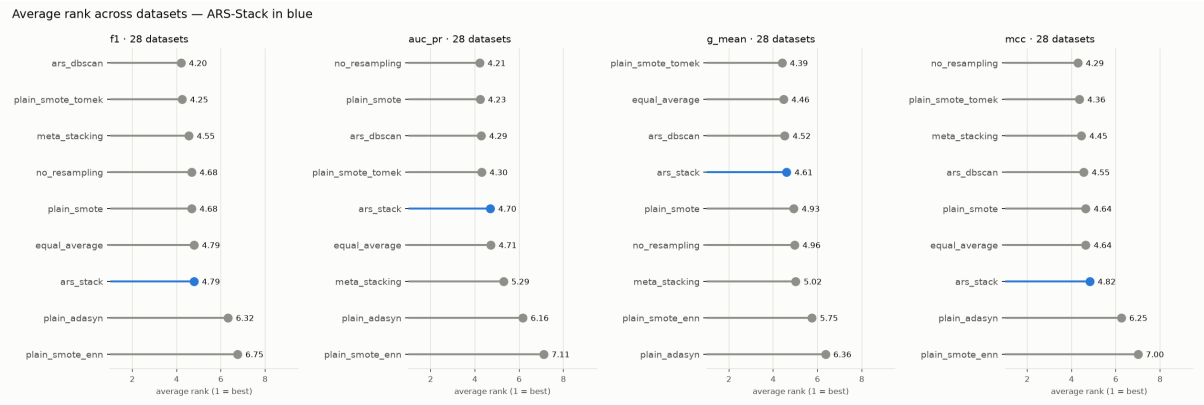


Figure 3: Average rank per metric over 28 datasets (1 = best; ARS-Stack highlighted). Apart from SMOTE-ENN and ADASYN, all variants lie within about one rank of each other.

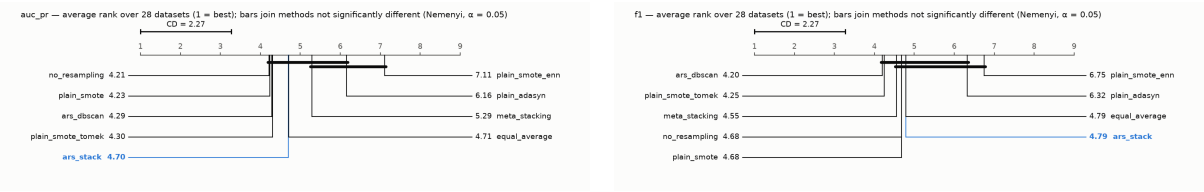


Figure 4: Critical-difference diagrams for AUC-PR (left) and F1 (right). Methods joined by a bar are not significantly different (Nemenyi, $\alpha = 0.05$, CD = 2.27).

Pairwise comparisons. Holm-corrected Wilcoxon tests on per-dataset means (Table 4) show ARS-Stack significantly better than SMOTE-ENN on F1 (21 wins/7 losses, $p_{\text{Holm}} = 0.041$), AUC-PR (22/6, $p = 0.049$) and MCC (22/6, $p = 0.041$). Its edge over ADASYN (19/9) does not survive correction ($p = 0.089$ on F1). Against all other variants the win/loss counts are near even, and the mean AUC-PR differences are small: -0.0045 against SMOTE, -0.0062 against No resampling, and $+0.0001$ against equal weighting.

Table 4: ARS-Stack against each variant over 28 datasets: wins/ties/losses of ARS-Stack, mean difference Δ (ARS-Stack − baseline) and Holm-corrected Wilcoxon p (bold: $p < 0.05$).

	F1			AUC-PR			MCC		
ARS-Stack vs.	W/T/L	Δ	p_{Holm}	W/T/L	Δ	p_{Holm}	W/T/L	Δ	p_{Holm}
Equal-weight avg.	11/8/9	+0.0003	1	12/3/13	+0.0001	1	9/8/11	+0.0003	1
Meta-learner stacking	13/3/12	+0.0011	1	16/1/11	+0.0019	0.983	12/3/13	+0.0005	1
ARS (DBSCAN)	10/5/13	+0.0017	1	9/5/14	-0.0011	1	12/5/11	+0.0024	1
SMOTE	11/0/17	+0.0011	1	12/0/16	-0.0045	0.879	11/0/17	+0.0025	1
SMOTE-Tomek	11/0/17	+0.0003	1	14/0/14	-0.0041	0.983	11/0/17	+0.0012	1
ADASYN	19/0/9	+0.0088	0.089	19/0/9	+0.0014	0.983	19/0/9	+0.0079	0.117
SMOTE-ENN	21/0/7	+0.0216	**0.041**	22/0/6	+0.0176	**0.048**	22/0/6	+0.0222	**0.041**
No resampling	14/0/14	+0.0066	1	12/0/16	-0.0062	1	13/0/15	+0.0025	1

Bayesian analysis. The Bayesian signed-rank test (Table 5) can support equivalence. It shows:

- **Practically better:** ARS-Stack beats SMOTE-ENN on F1 ($P = 0.96$), AUC-PR (0.98) and MCC (0.99).
- **Practically equivalent** ($P(=) \geq 0.95$) to SMOTE, SMOTE-Tomek, ARS with DBSCAN, equal-weight averaging and meta-learner stacking on all three metrics, and to ADASYN on AUC-PR and MCC.
- **Undecided** against No resampling. Equivalence is the most probable outcome ($P(=) = 0.79$–0.85), and on AUC-PR No resampling is more likely better than worse (0.15 versus 0.003).

The per-dataset Bayesian correlated t-tests are mostly undecided with 15 folds. Where they decide, they confirm the same picture: 76 of 112 dataset–metric pairs are equivalent to equal-weight averaging, and all 12 decided pairs against SMOTE-ENN favour ARS-Stack.

Table 5: Bayesian signed-rank test over 28 datasets (ROPE ±0.01): posterior probability that the baseline is practically better / equivalent / worse than ARS-Stack; a verdict needs $P \geq 0.95$.

	F1		AUC-PR		MCC	
ARS-Stack vs.	$P(<)/P(=)/P(>)$	Verdict	$P(<)/P(=)/P(>)$	Verdict	$P(<)/P(=)/P(>)$	Verdict
Equal-weight avg.	0.00/1.00/0.00	equiv.	0.00/1.00/0.00	equiv.	0.00/1.00/0.00	equiv.
Meta-learner stacking	0.00/1.00/0.00	equiv.	0.00/1.00/0.00	equiv.	0.00/1.00/0.00	equiv.
ARS (DBSCAN)	0.00/1.00/0.00	equiv.	0.00/1.00/0.00	equiv.	0.00/1.00/0.00	equiv.
SMOTE	0.00/1.00/0.00	equiv.	0.01/0.99/0.00	equiv.	0.00/0.99/0.01	equiv.
SMOTE-Tomek	0.00/1.00/0.00	equiv.	0.00/1.00/0.00	equiv.	0.00/0.99/0.00	equiv.
ADASYN	0.00/0.95/0.05	undecided	0.00/1.00/0.00	equiv.	0.00/0.98/0.02	equiv.
SMOTE-ENN	0.00/0.04/0.96	ARS better	0.00/0.02/0.98	ARS better	0.00/0.01/0.99	ARS better
No resampling	0.03/0.79/0.18	undecided	0.15/0.85/0.00	undecided	0.10/0.82/0.08	undecided

Per-dataset results. Table 6 lists every dataset with its characteristics and mean AUC-PR, ordered by minority-class size. No resampling has the highest AUC-PR on 12 of the 28 datasets. On webpage, us_crime, scene and wine_quality, every resampling variant loses 0.03–0.09 AUC-PR against it. Resampling wins mostly on datasets with tiny minority classes (arrhythmia, oil, ozone_level, abalone_19). Saturated datasets (pen_digits, car_eval_4, letter_img, optical_digits) leave all variants within 0.002 of each other, except SMOTE-ENN on car_eval_4.

For the core datasets (Table 7), the variants on creditcard differ by less than a standard deviation, and No resampling is best on mammography for every metric except G-mean. On 10-class yeast, resampling gives the largest gains in the study: macro-F1 rises from 0.441 to 0.524 and G-mean from 0.642 to 0.708. SMOTE-ENN, by contrast, collapses MCC to 0.388.

Literature baselines. Reproducing the "SMOTE + learner" baselines of our review on a fixed split with a 0.5 threshold (Table 8) gives lower scores than published: creditcard F1 is 0.78–0.83 against the reported 0.89–0.915. Three leakage-free choices account for the gap: we remove duplicates, apply SMOTE only after splitting, and hold the test set out. Oversampling before the split, common in applied work, lets synthetic interpolations of test points leak into training. The comparison is indicative only, since the published splits differ.

6 Why Does Adaptive Selection Not Help?

What the selector chose. Over the 420 ARS-Stack folds, the minority classes produced 970 clusters. The resampler applied was SMOTE-Tomek for 33% of them, SMOTE for 30%, SMOTE-ENN for 29%, ADASYN for 6% and random oversampling for 2%. Figure 6 shows that the choice is effectively made per *dataset*. On 11 datasets, one resampler is used for every cluster in every fold. The clustering does split the minority class, into 1.8 clusters on average and into several clusters in 57% of folds, but clusters of the same dataset receive similar scores and hence the same rule. DBSCAN always returned a single cluster, yet ARS with DBSCAN matches the k-means version (F1 rank 4.20 versus 4.79). Per-cluster heterogeneity is not what drives performance.

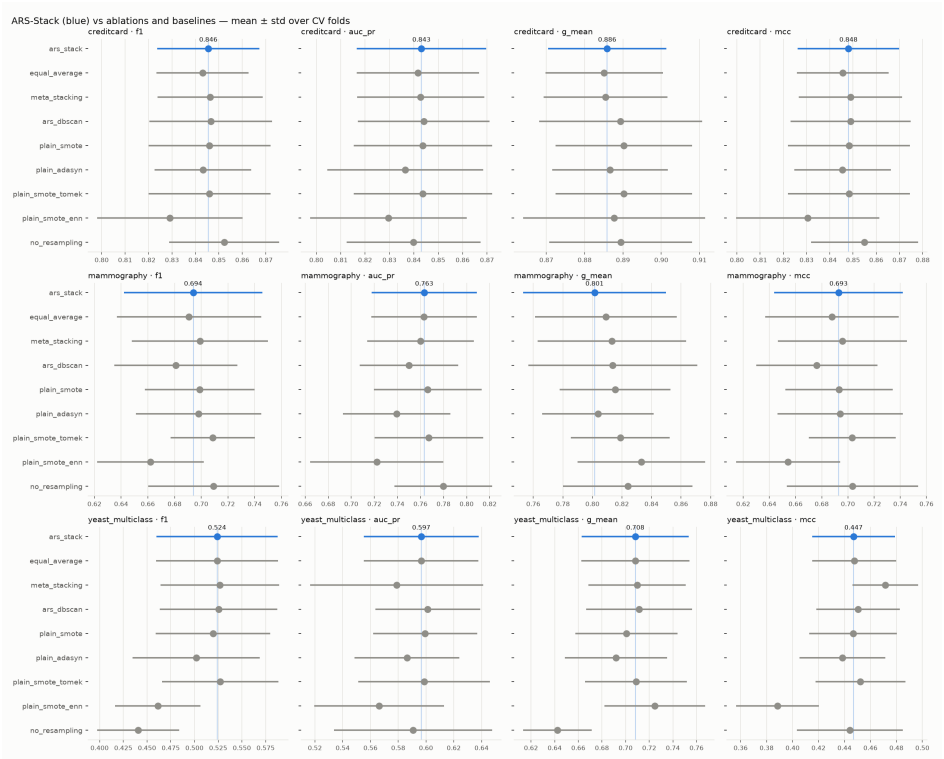


Figure 5: Core datasets: mean ± standard deviation over 15 folds for all nine variants and four metrics (ARS-Stack highlighted). Fold-to-fold spread exceeds most differences between variants.

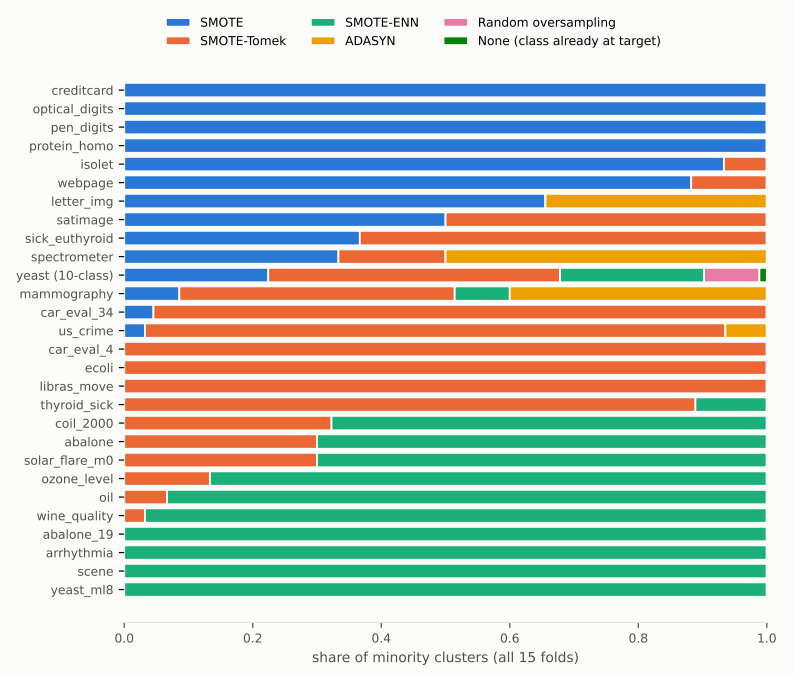


Figure 6: Resampler applied by ARS-Stack per minority cluster, as a share over the 15 folds of each dataset. Most datasets are dominated by one resampler.

Table 6: Datasets and mean AUC-PR over 15 folds (best per dataset in bold), ordered by minority-class size. n: samples after duplicate removal; d: features; Min.: minority (smallest-class) size; IR: imbalance ratio. Core datasets are marked *.

					Mean AUC-PR						
Dataset	n	d	Min.	IR	None	SMOTE	SM-Tomek	ADASYN	SM-ENN	Meta	ARS
yeast (10-class)*	1,453	8	5	87.6	0.591	**0.599**	0.599	0.586	0.566	0.579	0.597
libras_move	330	90	24	12.8	0.881	0.895	0.895	0.881	0.878	**0.899**	0.894
arrhythmia	452	278	25	17.1	0.815	0.833	0.833	0.837	0.783	0.840	**0.843**
abalone_19	4,177	10	32	129.5	0.049	0.068	**0.075**	0.056	0.056	0.058	0.055
ecoli	336	7	35	8.6	0.702	0.714	0.682	**0.719**	0.651	0.686	0.682
oil	937	49	41	21.9	0.580	**0.622**	**0.622**	0.620	0.573	0.603	0.600
spectrometer	531	93	45	10.8	**0.913**	0.878	0.885	0.854	0.858	0.865	0.865
solar_flare_m0	412	32	58	6.1	0.186	0.177	0.178	0.175	**0.246**	0.192	0.193
car_eval_4	1,728	21	65	25.6	0.999	1.000	1.000	1.000	0.972	**1.000**	**1.000**
ozone_level	2,536	72	73	33.7	0.321	0.369	0.369	0.361	**0.385**	0.355	0.356
car_eval_34	1,728	21	134	11.9	0.990	0.989	0.989	0.990	0.973	**0.990**	0.990
us_crime	1,994	100	150	12.3	**0.594**	0.546	0.548	0.546	0.523	0.549	0.550
scene	2,400	294	173	12.9	**0.271**	0.242	0.245	0.239	0.210	0.223	0.226
wine_quality	3,961	11	173	21.9	**0.322**	0.282	0.280	0.274	0.271	0.251	0.249
yeast_ml8	2,417	103	178	12.6	**0.108**	0.104	0.104	0.103	0.105	0.108	0.107
thyroid_sick	3,771	52	231	15.3	**0.954**	0.946	0.947	0.947	0.893	0.944	0.946
mammography*	7,849	6	254	29.9	**0.780**	0.766	0.767	0.739	0.722	0.760	0.763
sick_euthyroid	3,109	42	285	9.9	**0.898**	0.897	0.897	0.896	0.868	0.897	0.896
abalone	4,177	10	391	9.7	0.297	0.302	0.305	0.298	**0.317**	0.291	0.290
creditcard*	283,726	30	473	598.8	0.840	**0.844**	**0.844**	0.836	0.830	0.843	0.843
optical_digits	5,620	64	554	9.1	**0.995**	0.995	0.995	0.994	0.994	0.995	0.995
coil_2000	8,380	85	568	13.8	0.126	0.125	0.124	0.124	**0.140**	0.119	0.124
isolet	7,797	617	600	12.0	**0.967**	0.962	0.962	0.960	0.867	0.959	0.962
satimage	6,435	36	626	9.3	0.809	**0.811**	0.811	0.805	0.728	0.806	0.811
letter_img	18,668	16	640	28.2	**0.996**	0.995	0.995	0.995	0.995	0.995	0.995
webpage	34,780	300	981	34.5	**0.855**	0.814	0.814	0.788	0.761	0.788	0.814
pen_digits	10,992	16	1,055	9.4	0.998	0.998	0.998	0.997	0.998	**0.999**	**0.999**
protein_homo	144,968	74	1,251	114.9	0.885	0.901	0.901	0.888	0.893	0.901	**0.902**
Mean					0.669	0.667	0.667	0.661	0.645	0.660	0.662

Table 7: Core datasets: mean ± standard deviation over 15 folds (best in bold).

Variant	F1	AUC-PR	G-mean	MCC
creditcard				
No resampling	**0.852 ± 0.023**	0.840 ± 0.027	0.889 ± 0.019	**0.855 ± 0.023**
SMOTE	0.846 ± 0.026	0.844 ± 0.028	**0.890 ± 0.018**	0.848 ± 0.026
SMOTE-ENN	0.829 ± 0.031	0.830 ± 0.032	0.888 ± 0.024	0.830 ± 0.031
Meta-learner stacking	0.846 ± 0.022	0.843 ± 0.026	0.885 ± 0.016	0.849 ± 0.022
ARS-Stack	0.846 ± 0.022	0.843 ± 0.027	0.886 ± 0.015	0.848 ± 0.022
mammography				
No resampling	**0.709 ± 0.049**	**0.780 ± 0.042**	0.824 ± 0.044	**0.703 ± 0.050**
SMOTE	0.699 ± 0.041	0.766 ± 0.047	0.815 ± 0.038	0.693 ± 0.041
SMOTE-ENN	0.662 ± 0.040	0.722 ± 0.058	**0.833 ± 0.043**	0.654 ± 0.040
Meta-learner stacking	0.699 ± 0.051	0.760 ± 0.046	0.813 ± 0.050	0.696 ± 0.049
ARS-Stack	0.694 ± 0.052	0.763 ± 0.046	0.801 ± 0.048	0.693 ± 0.049
yeast (10-class)				
No resampling	0.441 ± 0.043	0.591 ± 0.057	0.642 ± 0.029	0.444 ± 0.041
SMOTE	0.520 ± 0.060	0.599 ± 0.038	0.701 ± 0.043	0.447 ± 0.034
SMOTE-ENN	0.461 ± 0.045	0.566 ± 0.047	**0.725 ± 0.043**	0.388 ± 0.032
Meta-learner stacking	0.527 ± 0.062	0.579 ± 0.062	0.710 ± 0.041	**0.471 ± 0.025**
ARS-Stack	0.524 ± 0.064	0.597 ± 0.041	0.708 ± 0.045	0.447 ± 0.032

SMOTE-Tomek is SMOTE. After 1:1 SMOTE, synthetic points fill the minority region so densely that almost no cross-class nearest-neighbour pairs (Tomek links) remain. Global SMOTE-Tomek gave *bit-identical* results to SMOTE on 280 of 420 folds, and on all folds of 12 datasets, while its cleaning step cost up to 114 s per creditcard fold. ARS-Stack's most frequent choice is therefore effectively SMOTE, and its real candidate pool is dominated by SMOTE.

Confidence weighting is equal weighting. The three learners reach similar validation AUC-PR, so the weights of Eq. 3 stay near 1/3. The median gap between the largest and smallest weight is 0.022

Table 8: Literature baselines on a fixed split (0.5 threshold) versus reported values.

Dataset	Model	Source	F1	AUC-PR	MCC
creditcard	RandomForest + SMOTE	ours	0.832	0.821	0.836
creditcard	XGBoost + SMOTE	ours	0.775	0.818	0.775
creditcard	Simple Stacking (RF+XGB, LR meta) + SMOTE	ours	0.815	0.823	0.815
creditcard	Paper 7: xgboost	reported	0.890	0.930	–
creditcard	Paper 7: easyensemble	reported	0.870	–	–
creditcard	Paper 9: stacking	reported	–	–	–
creditcard	Paper 10: stacking smote tomek	reported	0.915	0.900	–
mammography	RandomForest + SMOTE	ours	0.615	0.606	0.602
mammography	XGBoost + SMOTE	ours	0.630	0.639	0.618
mammography	Simple Stacking (RF+XGB, LR meta) + SMOTE	ours	0.647	0.636	0.635

(90th percentile 0.10). ARS-Stack and equal-weight averaging give identical metrics on 159 of 420 folds (mean AUC-PR difference +0.0001). The weights have nothing to correct when the learners are this strong and this similar. Meta-learner stacking is also equivalent, at 4.6× the compute (5.7 versus 1.2 hours).

When does resampling help at all? Figure 7 plots, per dataset, the gain from resampling over the same ensemble without resampling, against the minority-class size:

- ≤ 75 **minority samples (10 datasets):** SMOTE improves AUC-PR on 8 (mean +0.012); ARS-Stack on 8 (mean +0.005).
- > 75 **minority samples (18 datasets):** SMOTE improves AUC-PR on only 5 (mean −0.009); ARS-Stack on 4 (mean −0.012).

F1 and MCC show the same pattern. The Spearman correlations between gain and minority size are negative throughout (e.g. $\rho = -0.48$, $p = 0.010$ for ARS-Stack on F1), but none survives Holm correction over the 24 tests we ran. The imbalance ratio itself is not predictive. The absolute number of minority examples, not their proportion, appears to limit a strong learner. We report this as a consistent trend, not a confirmed effect.

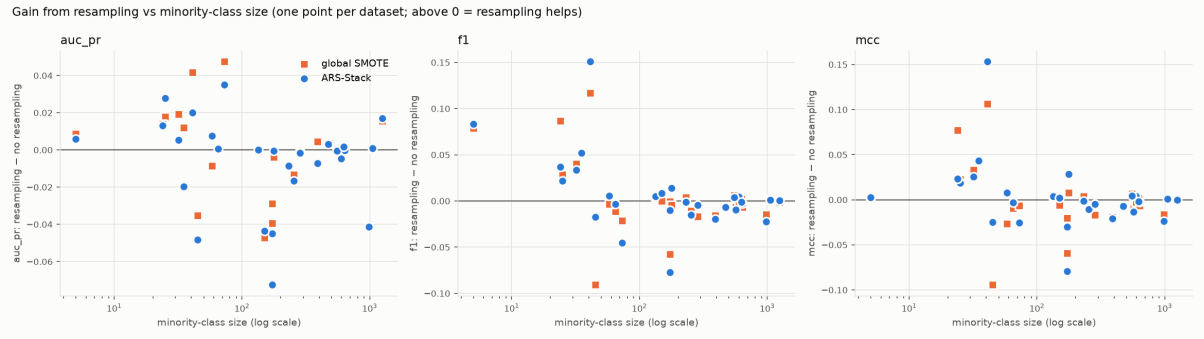


Figure 7: Gain from resampling (global SMOTE, ARS-Stack) over no resampling versus minority-class size (log scale), one point per dataset. Resampling tends to help below about 75 minority samples and to be neutral or harmful above.

Pilot: validated selection. Perhaps hand-written rules are just a poor selector. To test this, we replaced them with *validated selection*: the method (no resampling, global SMOTE, or ARS-Stack's adaptive resampling) and the amount ($\beta \in \{0.1, 0.25, 0.5, 1\}$) are chosen jointly by 3-fold inner cross-validation on the training data (LightGBM AUC-PR), and a single cluster is kept unless the silhouette score exceeds 0.25. A control, SMOTE-Select, runs the same search without the adaptive option.

To avoid tuning on the reported benchmark, the datasets were split into 14 development and 14 held-out sets, stratified by minority size and imbalance. The pilot used 12 development datasets and 5 folds, all on one machine. That matters: repeating an identical configuration on different hardware moved AUC-PR by 0.007 on average, as much as the effects studied.

Table 9 shows that validated selection repairs ARS-Stack's deficit (+0.0095 AUC-PR over the rule-based version) but only reaches the baselines: +0.0006 over No resampling (5 wins, 7 losses) and identical to SMOTE-Select. The adaptive option adds nothing, and the chosen configurations were spread over all candidates, a sign that they perform alike. The headroom is small in any case: an oracle picking the best of No resampling, SMOTE, SMOTE-Tomek and ARS-Stack per dataset in hindsight gains only +0.008 mean AUC-PR over No resampling. We therefore did not spend a second full run on this design.

Table 9: Validated-selection pilot on 12 development datasets (5 folds, one machine): mean score (average rank among the five variants).

Variant	F1	AUC-PR	G-mean	MCC
No resampling	0.4919 (2.92)	0.5301 (2.67)	0.6944 (2.92)	0.4696 (2.92)
SMOTE	0.4854 (3.67)	0.5290 (3.08)	0.6810 (3.75)	0.4602 (3.75)
ARS-Stack	0.4920 (2.92)	0.5212 (3.50)	0.6937 (3.08)	0.4651 (3.08)
SMOTE-Select (validated)	0.4951 (2.71)	0.5307 (2.83)	0.6989 (2.46)	0.4705 (2.62)
ARS-Select (validated)	0.4921 (2.79)	0.5307 (2.92)	0.6907 (2.79)	0.4669 (2.62)

7 Discussion

Implications. For strong tree ensembles with a tuned threshold, *which* oversampler is applied, and *where*, matters little (gaps G1–G2). The exception is aggressive cleaning: SMOTE-ENN, and to a lesser degree ADASYN, reliably hurt. Two mechanisms explain this. First, a strong learner combined with a tuned threshold already makes the precision–recall trade-off that resampling targets [27, 28]. Second, once budgets are matched and leakage removed, most resamplers produce near-identical training sets. Gap G3 fares better: multiclass problems and datasets with a few dozen minority examples show the largest gains, and resampling research should focus there.

Five evaluation pitfalls. Our first pipeline contained five errors, each of which would have produced a spurious improvement:

1. **Unequal budgets.** Local balancing added far less synthetic data than global SMOTE (a 1:200 versus 1:1 final ratio on creditcard), confounding resampler choice with resampling amount.
2. **Scale-cancelled weights.** Confidence weights fed into a meta-learner are a no-op.
3. **Rank-forced rules.** Percentile rules with one or two clusters decide by rank.
4. **Fixed 0.5 thresholds.** These make F1, G-mean and MCC mirror the oversampling ratio.
5. **Duplicates and pre-split resampling.** Both place near-copies of test points in the training data.

Controlling for all five is, in our view, a minimum standard for resampling papers.

Threats to validity.

- **Fixed hyperparameters.** Hyperparameters and rule thresholds were fixed a priori, not tuned per dataset. Tuning would affect all variants alike.
- **Hardware non-determinism.** It exists (GPU versus CPU XGBoost), so the study ran on one machine from one code version.
- **Small validation splits.** The 15% split is small for tiny minority classes.
- **External validity.** The benchmarks form a single collection. Deduplication limits comparability with prior work.
- **Learner scope.** The conclusions concern strong tree ensembles; weaker learners may benefit more from resampling [29].
- **Statistical power.** Per-dataset power is limited by 15 correlated folds, and the minority-size trend does not survive multiplicity correction.

8 Future Work

- **Other learners.** Repeat the study with logistic regression, k-NN, SVMs and deep tabular models, to map the learner–resampler interaction.
- **The small-sample regime.** Test the minority-size trend in a pre-registered study on datasets with fewer than 100 minority examples, including multiclass ones, and add generative oversamplers.
- **Meta-learned selection.** Replace rules and noisy inner validation with a meta-model that predicts the benefit of resampling from dataset characteristics.
- **Combination.** Replace global confidence weights with dynamic ensemble selection [38].
- **Alternatives to resampling.** Compare resampling with cost-sensitive learning, and evaluate calibration [28].
- **Analysis and deployment.** Use hierarchical Bayesian models across datasets [25], and add drift monitoring and energy accounting [13] to the served models.

9 Conclusion

We proposed ARS-Stack, which selects a resampler per minority cluster from overlap and density scores and combines three tree ensembles with confidence weights, and we evaluated it in a budget-matched, leakage-free study on 28 datasets.

ARS-Stack reliably beats aggressive cleaning (SMOTE-ENN). It is practically equivalent to plain SMOTE and to its own ablations, and it cannot be separated from training without resampling. Each null result has a concrete cause: near-uniform weights, cleaning that removes nothing after balancing, and selection that rarely varies within a dataset. A validated-selection pilot confirms that the headroom is small, and resampling mainly pays off when minority examples are scarce.

We hope the method, the pipeline and the five pitfalls help future work target the settings where resampling matters, and measure its benefit fairly.

Availability. Code, configuration, per-fold results and analysis scripts are in the project repository. All datasets are public: Kaggle [58], imbalanced-learn [57] and UCI.

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